Auteur: Torben Petré
Studierichting: Informatica
Oefeningenreeks 1c nr. 44.4

Trek op meetkundige manier de vierkanstwortel in $\mathbb{C}$:

$z^2 = -i$

$$\begin{cases} r = \sqrt{0^2 + (-1)^2} \\ \tan\theta = -\dfrac{\pi}{2} \end{cases}$$

$n = 2$

- $w_0 = \operatorname{cis}\left(-\dfrac{\pi}{4}\right)$
  $= -\dfrac{\sqrt{2}}{2} + \dfrac{\sqrt{2}}{2}i$

- $w_1 = \operatorname{cis}\left(\dfrac{-\dfrac{\pi}{2} + 2\pi}{2}\right)$
  $= \operatorname{cis}\left(\dfrac{\dfrac{3\pi}{2}}{2}\right)$
  $= \operatorname{cis}\left(\dfrac{3\pi}{4}\right)$
  $= \dfrac{\sqrt{2}}{2} - \dfrac{\sqrt{2}}{2}i$

References